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Data Acquisition and Processing I: Big Data

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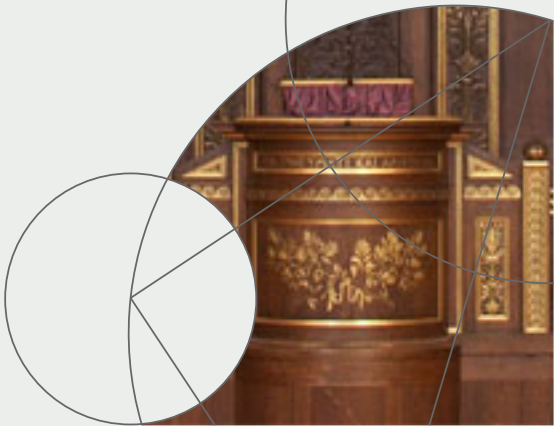


conditions for AR(p) stationarity

- More generally for an AR(p) process, we can define the characteristic equation

$$z^p - a_1 * z^{p-1} - \cdots - a_p = 0$$

- The AR(p) process will be stationary if and only if the roots of this characteristic equation are all less than one in absolute value; i.e. they lie inside the unit circle on the complex plane.
- Note that an AR(1) process is essentially a Markov process:

$$P[X_n = v_n | X_{n-1} = v_{n-1}, \cdots, X_0 = v_0] = P[X_n = v_n | X_{n-1} = v_{n-1}]$$

- A generalized definition of a Markov process is that the current distribution depends only on a fixed finite number p of previous observations, not the entire history. Thus an AR(p) process fits this more general definition of a Markov process.

Time series: Yule-Walker equations

- Assume for simplicity that the parameter c in an AR(p) process is zero and therefore all means are zero. Multiply the defining AR(p) equation by X_{t-k} ($k > 0$) and take expectations to get

$$E[X_t X_{t-k}] = \sum_{i=1}^{p} a_i E[X_{t-i} X_{t-k}]$$

- If the process is stationary, only the difference in times matters in defining autocovariances, so we have

$$v[k] = \sum_{i=1}^{p} a_i v(k-i) \quad \text{and} \quad v(0) = \sum_{i=1}^{p} a_i v(-i) + \sigma^2$$

- Dividing both sides of the left-hand equation by $v(0)$ converts the v'âĂŹs (autocovariances) to râĂŹ's (autocorrelations) and gives rise to a series of linear equations called the Yule-Walker equations (noting $r(-i) = r(i)$):

$$\begin{pmatrix} r(1) \\ r(2) \\ \vdots \\ r(p) \end{pmatrix} = \begin{pmatrix} 1 & r(1) & \cdots & r(p-1) \\ r(1) & 1 & \cdots & r(p-2) \\ \vdots & \vdots & \vdots & \vdots \\ r(p-1) & r(p-2) & \cdots & 1 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_p \end{pmatrix}$$

MA processes, invertibility

- A moving average process MA(q) is one where

$$X_t = c + \epsilon_t - b_1 * \epsilon_{t-1} - \cdots - b_q * \epsilon_{t-q}$$

- We showed above that a stationary AR(1) process can be transformed into an infinite MA process going back to 0. In general if the AR(p) characteristic equation has roots inside the unit circle, the (therefore) stationary process can be transformed into such an infinite MA process.
- If we define the characteristic polynomial for an MA(q) process

$$z_q - b_1 * z_{q-1} - \cdots - b_q = 0$$

- Then if its roots are inside the unit circle (absolute value less than one), we can invert the MA(q) process into an AR process. This AR process will be stationary.

ARMA processes

- Combining AR and MA processes, we can define an ARMA(p,q) process as one where

$$X_t = c + a_1 * X_{t-1} + \cdots + a_p * X_{t-p} + \epsilon_t - b_1 * \epsilon_{t-1} - \cdots - b_q * \epsilon_{t-q}$$

- It is convenient to introduce the backward and forward operators B and F. For any time series Y_t, $BY_t = Y_{t-1}$, and $FY_t = Y_{t+1}$. B is sometimes called the lag operator.
- We can therefore rewrite an ARMA(p,q) process as

$$X_t = c + \left(\sum_{i=1}^{p} a_i B^i\right) X_t + \epsilon_t - \left(\sum_{i=1}^{q} b_i B^i\right) \epsilon_t$$

- Assuming unit roots of the AR characteristic equation, we can invert:

$$X_t = \frac{c}{1 - \left(\sum_{i=1}^{p} a_i\right)} + \frac{\left(1 - \sum_{i=1}^{q} b_i B^i\right)}{\left(1 - \sum_{i=1}^{p} a_i B^i\right)} \epsilon_t$$

ARMA processes

- Or (assuming unit roots of the MA characteristic equation) we can invert the other way:

$$\epsilon_t = \frac{-c}{1-\left(\sum_{i=1}^{q} b_i\right)} + \frac{\left(1-\sum_{i=1}^{p} a_i B^i\right)}{\left(1-\sum_{i=1}^{q} b_i B^i\right)} X_t$$

- Generally working with the backward operator polynomial in the denominator is messy, but it makes for a clean statement. This allows us to extend easily the definition of a stationary ARMA process to an ARIMA (Autoregressive Integrated Moving Average) process. An ARIMA(p,d,q) process can be written

$$\left(1-\sum_{i=1}^{p} a_i B^i\right)(1-B)^d X_t = c + \left(1-\sum_{i=1}^{q} b_i B^i\right)\epsilon_t$$

- where if we remove the $(1-B)^d$ term, the two backward operator polynomial characteristic functions have roots inside the unit circle and the remaining process is stationary ARMA(p,q). d is the level of differencing âĂŞ e.g. $d=1$ means we replace X_t with $Y_t = (X_t - X_{t-1})$ and have an ARMA(p,q) process in first differences. We impose $p+d$ initial conditions on X, and q initial conditions on ϵ.

Parameter estimation

- Having defined various kinds of processes, we now proceed to look at techniques to determine the kind of process observed values may originate from. There are many books on this process. We will give a few of the major techniques. We start with WoldâĂŹs Decomposition Theorem:
  - Every zero mean covariance stationary time series can be represented in the form

$$X_t = E\left[X_t | X_{t-1}, \cdots, X_{t-p}\right] + \sum_{i=0}^{\infty} w_i \epsilon_{t-i}$$

  - where the ϵ's are i.i.d. normal, $w_0 = 1$, and the sum of the absolute values of the w_i are finite. $E[X_t | X_{t-1}, \cdots, X_{t-p}]$ is the regression of X_t on the lagged values, and is uncorrelated with the ϵ's.
- This suggests doing regression to determine lags. We use this together with the Box-Jenkins approach, which fills a book but which can be summarized as:
  - Step 1. Transform the time series data so it looks stationary. This can be done by taking differences (ARIMA to ARMA); taking logs or other functional transforms; and removing trends. The goal is to have the autocorrelations drop off sharply after a certain value.

Parameter estimation

- Step 2. Estimate the parameters of the transformed data as an ARMA model, using WoldâĂŹs theorem âĂŞ i.e. least squares regression. We can use the Yule-Walker equations as a short cut for regression.
- Step 3. Generate the residuals from the estimated model and use a statistical test to see if they look like uncorrelated normal variates âĂŞ white noise. If not, start over.

- Assume then that we have data $Y(1), \cdots, Y(T)$. We form $m(Y)$, the sample mean of Y, from all T observations. We then compose the sample autocorrelations

$$sr_k = \frac{\sum_{t=k+1}^{T} [Y(t) - m(Y)][Y(t-k) - m(Y)]}{\sum_{t=1}^{T} [Y(t) - m(Y)]^2}$$

- We plot the correlogram with k on the horizontal axis and sr_k on the vertical. A sharp drop above some value of k may indicate the degree of autocorrelation of the process.

Parameter estimation

- Let us try an AR(2). The Yule-Walker equations for $p = 2$ look like:

$$\begin{bmatrix} r(1) \\ r(2) \end{bmatrix} = \begin{bmatrix} 1 & r(1) \\ r(1) & 1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$$

- Both sides multiplying with inverse of the matrix $\begin{bmatrix} 1 & r(1) \\ r(1) & 1 \end{bmatrix}$

$$\frac{1}{1 - r^2(1)} \begin{bmatrix} 1 & -r(1) \\ -r(1) & 1 \end{bmatrix} \begin{bmatrix} r(1) \\ r(2) \end{bmatrix} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$$

$$a_1 = \frac{r(1)(1 - r(2))}{1 - r^2(1)}; \qquad a_2 = \frac{r(2) - r^2(1)}{1 - r^2(1)}$$

- For example, from our correlogram we have $r(1) = .966554$ and $r(2) = .908977$. This gives the predictive model

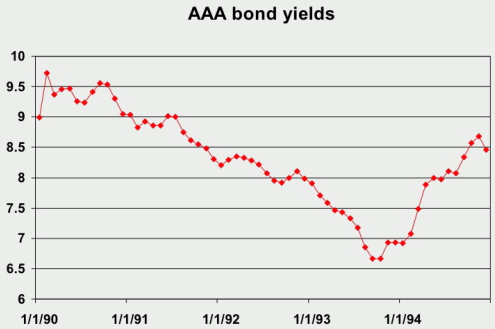
$$X_t = c + 1.337606 * X_{t-1} - 0.38389 * X_{t-2} + \epsilon_t$$

Parameter estimation

- We use the Lyung-Box statistic to test the hypothesis that, given p sample autocorrelations, the series is just white noise (no autocorrelation):

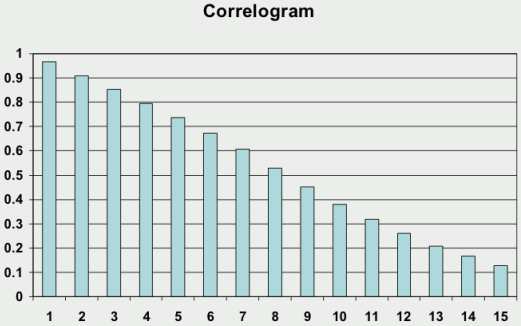
$$\mathrm{LB} = T(T+2)\sum_{k=1}^{p}\frac{sr_k^2}{T-k}$$

- This is distributed as χ^2 with p degrees of freedom. We will now use these techniques on an example. Here are five years of monthly observations of yields on AAA bonds âĂŞ these are the highest rated bonds.



Parameter estimation

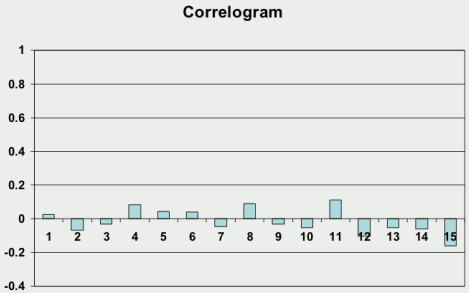
- Our first observation is that, given the big dip in late 1993-early 1994, this may not be a stationary process. However, that could be random variation.
- We will form the correlogram to see if there are obvious lag effects.



- The Ljung-Box statistic on the first 15 autocorrelations is 360.57, which with a chi-square with 15 degrees of freedom rejects randomness with a p-value that is very close to zero. This is clear anyway from the high autocorrelations. The partial correlogram indicates that an AR(1) or AR(2) process might be reasonable, although the correlogram doesnâĂŹt drop off sharply the way we would expect in a real AR(2) process.

Parameter estimation

- We can show that $c = E[X_t] * (1 - a_1 - a_2)$. Substituting sample values gives $c = .383748$. Check the characteristic equation $z^2 - 1.34z + .38 = 0$. Roots are -0.417 and -0.920, inside the unit circle as desired. This process is stationary.
- We now analyze the residuals formed by subtracting the predictive equation from the original data. The residual correlogram looks like



- The Ljung-Box statistic on the first 15 autocorrelations is now 6.79, which fails to reject randomness with a p-value of 96.3%.